AWI–ESM3 v3.4 (TCO95L91 / CORE3) — tuning parameter summary

Campaign 2026-06-30 to 2026-08-24 · current best coupled arm: **11R** (LX4 stack + `RSBLB` 2.0, 1990 forcing) / **11Q** (1850)

1. Kept — the adopted stack as it stands in 11Q / 11R

Everything below is live in the current best arm. Defaults are the as-released values.

round	parameter	default	adopted	what it does / why kept
OpenIFS — cloud and radiation				
D2b	`RCL_INPSEA`	1.0	**0.2**	Marine ice-nuclei floor over ocean. Southern-Ocean cloud phase. Spent: phase is already more liquid than CALIPSO-GOCCP at 45–65S
S4	`RCL_INPPMIN`	70000	**50000**	**UNDER REVIEW** — INP gate depth. Buys nothing at the pole, authored the −3.70 sub-Antarctic overcorrection, and its Siberian JJA benefit reverses coupled (+1.19 AMIP → −0.37). Arms 11V/11W testing removal
B7⁻	`RVICE`	0.13	**0.16**	Ice fall speed
LX4	`RSNOWLIN2`	0.035	**0.04**	Snow autoconversion. The only lever in ∼52 AMIP arms that moves the tropical longwave
32	`RSBLB` *(new)*	5.0	**2.0**	Stable-BL stability-function coefficient b, exposed as `&NAMVDFS` this campaign. The winter land lever: Arctic DJF screen +0.67 K coupled, zero Siberian summer cost, zero net TOA
OpenIFS — land surface				
G4	`ECE_TUNE_RVRSMIN(3,4)`	250	**1000**	Stomatal resistance, needleleaf. Less transpiration → more sensible heat
G4	`ECE_TUNE_RVRSMIN(9)`	80	**225**	Same, tundra — 25.6 % of the boreal box that F4 missed
P3	`ECE_SNOW_SCF`	1	**3**	Snow-cover depletion fitted to observations, resolution-aware
P3	`..._RHOREF/DCL/DCH/MD/BL/BH/DCMAX/SCALE/SWEMIN`		9 values	The fitted depletion curve: 200, 0.014, 0.026, 4.70, 1.46, 0.40, 0.30, 1.0, 15.0
K1	`ECE_TUNE_RVVEGALB`	—	**12 values**	Vegetation albedo, types 1/10/11 only, all four spectral components
K1	`ECE_TUNE_SOILALB(1-4)`	1.0	**0.95**	Bare-soil background scale
—	`LRDALB`	True	**False**	Coupled albedo on, LPJG-driven
OpenIFS — other				
09C	`ENTSTPC3`	3.0	**1.0**	Extra entrainment, PBL inversion height only
—	`GGAUSSB`	—	−0.5	Non-orographic gravity-wave source
26	`ECE_CLIMR_DMS`	.false.	**.true.**	DMS-Rev3 ICMCL field READ . . .
26	`ECE_DMS_CCN_SENS`	0.0	**0.0**	. . . but the CCN lever left **OFF**: at S=166 it costs −2.35 W m^{-2} in the tropics and is not polar-selective
FESOM / sea ice — inherited from the 06T/09B ocean round				
09B	`pndaspect`	0.8	**1.3**	Melt-pond aspect ratio
09B	`rfracmax`	1.0	**0.75**	Max melt-pond refreeze fraction
09B	`albpnd`	0.2	**0.28**	Melt-pond albedo
09B	`use_momix` / `momix_kv`	.false.	**.true.** / 0.01	Monin–Obukhov mixing
09B	`kpp_av0` / `kpp_kv0`	0.005	**0.003**	KPP mixing. Suppresses Weddell/Ross open-ocean convection; improves 500–1000 m by ∼0.15 K
09B	`kpp_avbckg`	1e-4	**5e-5**	KPP background momentum diffusivity
09B	`kpp_kvbckg`	1e-5	**5e-6**	KPP background tracer diffusivity
—	`spp`	—	**false**	Stochastic physics off
Coupling				
06T	`oasis3mct.time_step`	7200	**3600**	1-hour coupling

2. Tried and discarded

Struck through where the mechanism itself was falsified rather than the arm merely underperforming. Red marks a verdict overturned or reopened by later work.

arm	parameter	change	why discarded
Cloud overlap, ice nuclei, microphysics			
A1a	`RCL_OVERLAPLIQICE`	0.65→0.10	Overshoots: SO CRE −6.51 but Siberian JJA −1.15 K. Wrecks the boreal
A1b	`RCL_OVERLAPLIQICE`	0.65→0.35	Best early arm, superseded; coupled as 11L it costs 1.000 K of Siberian JJA
11L/11M	`RCL_OVERLAPLIQICE`	0.35 / 0.10	Coupled. First SO lever that transfers (0.57–1.01×), both rejected: 11M fails net TOA, 11L spends the whole Siberia budget for 40 % of the SO target
A1c	`RDEPLIQREFDEPTH`	500→1500	**Reopened.** Recorded "failed — cloud-depth selectivity idea dead" on the 45–65S metric. Re-scored on the polar band it is significant (−1.30 ocean DJF) and nearly free (TOA +0.01)
A2	`RCL_KK_CLOUD_NUM_LAND`	300→150	Nothing significant anywhere
B1	`DETRPEN`	0.75e-4→0.45e-4	More SW yet colder; worst tropics and global RMSE
B2	`RCLDIFF_CONVI`	10→25	**Reopened.** Killed on subpolar N. Atlantic RMSE. On the polar band it is the *only* arm passing both vetoes: polar −0.68, Arctic +0.13 (null), sub-Antarctic +0.18, Siberia +0.38 K
B3	`RCLDIFF`	6e-6→1.5e-5	Energy fails: surface flux +1.14, tropics +1.72
B4	`ENTSHALP`	2.0→3.0	Boreal within noise
B5	`RCAPDCYCL`	2.0→0.0	Boreal within noise
B6	`RLCRITSNOW`	→1e-5	No traction
B7	`RVICE`	→0.22	Superseded by 0.16
N1/N2	`RCL_INPSEA`	0.2→0.1 / 0.05	Phase is already *more* liquid than GOCCP at 45–65S; buying CRE by worsening phase
W1	~~`RCLCRIT_SEA`~~	2.5e-4→6e-4	**Dead parameter**: never read in the `IWARMRAIN==3` KK branch. 46 yr identical to control
SC2/SC3	`RDEPLIQREFRATE`	0.5→0.2	SC3 (rate+depth) breaks the Arctic (−1.85, already −18 overcorrected). Rate is what destroys selectivity, not depth. SC2 cancelled at 24 yr
26	`ECE_DMS_CCN_SENS`	0→166	Works at the pole (−6.10 DJF) but hits 60–45S equally (−6.55), so it deepens the overcorrection that blocks it. Tropics −2.35, global TOA −1.61
Land surface and boundary layer			
expA	`RVRSMIN(3,4)`	250→500	Within noise; sign unconfirmed
F1	`ECE_TUNE_RVZOH`	forest, /10	No boreal gain. Wrong side of the column: the skin bias is only +0.11 K
F2/F3	`RVLAI` / `RVCOV`	—	No significant boreal gain
F5	F1+F2+F3+F4	—	+0.746, no better than F4 alone (+0.749): near-total saturation
B8/E1	`ECE_TUNE_RVLAMSK`	→5 / 25	+0.502 at 4 yr was pure noise; −0.038 at 44 yr
J1/J2	~~`ECE_LAMSK_SN`~~	7→15 / 25	**Falsified.** Predicted +1 to +3 K DJF; measured +0.125 (thr 0.41) and −0.039. The skin is already right
C1/C2	`RLAM`	150→75 / 40	Null. Momentum mixing length, not heat
SB3	~~`RSBLLMIN`~~	30→120 m	**Falsified.** Same Arctic gain as `RSBLB` at 4× the tropical cost and +1.67 W m^{-2} net TOA: it also sets the length in free-shear layers, so it strips stratocumulus
D1	`RCAPDCYCL` mode 4	—	Mechanism falsified
Vegetation and coupling			
11H/11J	~~`ifraupachz0`~~	off→on	**Falsified twice.** Structurally (Raupach needs a canopy the model has lost: $h = 2.89$ m, $z_0 = 0.0132$ against IFS's 0.058) and by measurement (DJF land–ocean gap −0.017 and −0.003 in two windows)
06O/06T	`kpp*` + `mospp`, aggressive	—	Improves the ocean interior but "the cooling leaks well beyond where an ocean-mixing parameter has any business acting" — tropics and Niño-3.4 redden. Milder settings retained (§1)
06V	`ENTSTPC3`	aggressive	Largest atmospheric degradation: `tas` +0.45, `zg` +0.46

Standing caveats. (i) AMIP→coupled transfer has been wrong in *sign*: S4 gives +1.19 K of Siberian JJA in AMIP and −0.37 K coupled. (ii) Two `lpj_guess` binary generations exist (`production_8c5ab467` for 11E/11G, `raupach_junebase_gated` for 11L–11R); comparisons spanning them are confounded, and all arms are now pinned. (iii) The PI controls start near zero net TOA (-0.06 to -0.08) and *drift* to $\sim +1.0$ over 40 years, driven by SH sea-ice loss — the energy problem is a drift, not an offset.